THE SATURDAY 60 — System Manual

The Song BPM Selector, the MOWL-Style Build Sheet, the Drops Library Editor, the Spin Class Pro workbook, the 5-Minute FTP Test and RPE + Watts cards, and the Spotify → Exportify/Chosic → MOWL pipeline · rev. 46 · October 2026

The three HTML tools now run as one installed web app at `https://indoor-cycling.github.io/saturday60/`

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0 · THE PHILOSOPHY AND THE THREE RULES

Every tool is built around one idea: **PLAYLIST FIRST — MUSIC LEADS.** Song structure (drops, choruses, builds) dictates interval timing; the clock follows the music on work segments, and only the fixed pieces — warm-up, bridges, cool-down — hold clean clock time.

Beat-locked RPM	Legs ride the beat one of exactly two ways: **1:1** (RPM = BPM) or **½** (RPM = BPM ÷ 2). Nothing else. On import, anything over 110 BPM is set to ½ automatically — unless a per-song override says otherwise.
110 RPM ceiling	Every interval is capped at 110. Songs may be faster; legs never are.
Rider zones REV 24	Power and HR targets derive from your FTP and maximum heart rate, set in the Build Sheet's header line and stored on the device (Keiser M3 / M3i Studio Plus, 72 gears). The template ships with FTP 158 W and Max HR 164 bpm.

1 · THE PIECES

Song BPM Selector	Phone-first. Tap-tester, phrase strip, block matching, playlist, drop capture with colour coding, song length, drops library, transfer codes, backup.
Build Sheet Template	Desktop-first. The 60-minute plan: reorderable sections, auto RPM, clock math, ride-clock drops, drill presets, Section Library, exports.
Build Sheet · no targets REV 34	No longer a separate file. It is now the **Hide targets** switch on the Build Sheet itself: the same sheet with the Power target and HR guide columns hidden and that width given to Song and Intervals. One sheet, one set of songs, drops and drills — nothing to keep in step and no second storage area to drift.
Drops Library Editor	Standalone. Opens a drops .json to view or edit outside the two tools: times in m:ss, drills alongside, merge two libraries, save back out, and **Export CSV** for a printable copy. REV 21
Spin Class Pro workbook	Excel backup and visuals: My Playlist + Intervals → Charts (segment pie, cadence bands, two load curves) and a two-sided fold-up Playlist Print.

Weekly Workflow	The one-page checklist of the six phases — in the hub's **Documents** group, readable in the app with a link back, plus a print-ready PDF that opens in its own window. REV 21
Getting Started REV 37	A one-page sheet for handing the app to another instructor: the address, how to install it on a phone or a computer, what the three tools do, and the half-dozen things worth knowing on day one — chiefly that their data is local to their own device and starts empty. In the hub's **Documents** group, with a print-ready PDF and a **Copy the link to share** button that puts the app's address on the clipboard, ready to paste into a message. There is nothing to package or download: the web address *is* the app.
System Manual REV 25	This document, in the hub's **Documents** group — readable in the app, stored for offline use, with a link back and a print-ready PDF beside it. The repository holds both the HTML and the PDF; change the HTML and re-render the PDF from it so the pair never disagree.
External tools	Spotify, Chosic Playlist Analyzer, Spotlistr, MOWL Session Designer, Musixmatch / YouTube heatmap.

The three HTML tools are no longer loose files. They are published together as an installable web app — one icon, working offline, with a hub screen linking the Selector, the Build Sheet and the Drops Library Editor. Because they now share one web address, they also share one storage area: a drops library saved in the Selector is the same one the Build Sheet reads. Data still lives per device. See Section 9.

2 · THE SONG BPM SELECTOR

The gauge and the phrase strip

Type a BPM or nudge with the −5 / −1 / +1 / +5 steppers. Cards list every block that tempo fits with its beat-locked RPM; anything faster than the cap gets the half-time rescue note. Beneath the number, the **phrase strip** shows how long 1, 8, 16 and 32 bars run at that tempo — at 126 BPM a bar is 1.90 s and a 32-bar phrase is 61 s.

The bar calculator REV 27

Where	Inside the Song tempo box, under the 1 / 8 / 16 / 32-bar strip. Type any number of bars and it answers at the tempo showing above — change the BPM and the answer follows.
What it answers	The run time and the beat count: at 126 BPM, 12 bars reads **22.9s · 48 beats**. Under a minute it keeps one decimal, because half a second decides whether a cue lands on the phrase; past a minute it switches to m:ss.
Why it exists	The strip covers the phrase lengths that matter most, but real tracks hold 12-bar bridges, 6-bar builds and the odd half bar. This is for those — sizing a drill to a section you have counted by ear, or checking whether an effort will fit before the next chorus.
Shortcuts	Tap any figure in the strip above to load that bar count. Fractions like 8.5 are accepted. The × clears it. The count is remembered on the device between visits.
The arithmetic	One bar is 240 ÷ BPM seconds (4/4 assumed, as everywhere in these tools), so N bars is N × 240 ÷ BPM, and the beats are N × 4.

The tap-tester

Tap the TAP pad in time with the music (4+ taps; keep going to refine). A 2-second pause resets it. **Use** sends the measured BPM to the gauge and the playlist form. This is the referee whenever a database's BPM looks doubled or halved.

Playlist builder

Add title + BPM; songs file under their best block and move with the per-song block dropdown. **Copy list** exports everything grouped by block, drops and overrides included.

Removing a song REV 19	The × takes two taps. The first turns it into a red **Delete?**; the second removes the song. It disarms itself after four seconds, if you tap anything else, or if you arm a different song's ×. Only one is ever armed at a time.
Hover help REV 19	Every button in all three tools carries a plain-language tooltip describing what it does. These appear on hover, so they help on a computer; a phone has nothing to hover with.

Per-song ride mode — the 1:1 / ½ button

Each song has its own ride-mode button that cycles three ways: *follow the block* (the default, unstyled), **1:1** forced, **½** forced. An overridden song lights up in the block's colour and its line reads "set". The override travels in transfer codes and beats the Build Sheet's automatic over-110 rule, so a deliberate choice is never silently undone.

Song length REV 19

On each song's drop-marker screen, under the downbeat field, is a **song length** box in m:ss, with a **from timer** button that takes whatever the stopwatch is showing. It is optional — leave it blank and nothing changes.

Why it exists	Runtime used to live only on the Build Sheet, so importing a playlist could not fill it. A length set here travels with the song and lands in the sheet's Len column on import.
Where it shows	On the playlist row after the RPM — "126 BPM · 63 RPM · 3:42".
Where it travels	Transfer codes, backups, and Import ← Selector. Not stored in the drops library.
Bad input	Anything unreadable is refused and the previous value returns. Clearing the box removes the length.

Drop markers — the ◎ button

Capturing	Stopwatch (Start/Pause/Reset) plus a big Mark drop button; chips show what you caught, each removable. Add by hand takes one typed time; Paste a list takes the whole song at once.

Pasting a whole list REV 43	Under Add by hand is a **Paste a list** box that takes every drop for a song in one go. Commas, spaces and line breaks all separate, so "0:32, 1:14, 2:05", one-per-line, and a line copied off another screen ("◉ 0:32 · 1:14 · 2:05") all read the same. Each time goes through the same parser the single field uses, so 1:14 and 74 mean the same thing. Times come back sorted with repeats dropped. **Add all** keeps the drops already captured and adds the new ones; **Replace** clears the song's drops and uses the pasted times instead, asking first when the song already has some. A line under the box reports what happened — "4 added", "6 drops set · 2 cleared", "1 already there". Colour coding survives: a time that is in both the old and the pasted list keeps its colour, and a slot whose time disappears is cleared.
Five ways to capture REV 23	Split-screen with Spotify and mark live · split-screen with Musixmatch and tap when the lyrics flip to Chorus (the drop sheet fits a half-height window) · read Spotify's elapsed clock and type it (most accurate) · scout the YouTube most-replayed peak in advance · **hold a phrase boundary** in the grid and take the time straight off it.
Simultaneous start	Tap the song's restart control and the stopwatch's Start together on your own count — you are launching both from zero, not catching a moving song.
Phrase boundaries	An 8 / 16 / 32 toggle lists every boundary time in the song — the natural places a drill can start or end. On the 8-bar view every fourth marker is brightened because it is also a 32-bar boundary; any boundary near one of your drops is highlighted. The list runs eight phrases ahead of the downbeat and extends as you mark drops past it; switch to 32 bars to see further into the track at a glance.
Holding a boundary REV 23	Press and hold any time in the phrase grid for half a second and it becomes a captured drop — no stopwatch, no typing. A bar sweeps across the chip as you hold, the phone gives a short buzz when it takes, and the boundary turns green to show it is now one of your drops. Hold a green one and it comes back out again, so the same gesture is its own undo. A short tap does nothing and sliding your finger cancels, so scrolling the grid never captures by accident. This is the fastest route when you already know the structure and are choosing where a drill should start.
Snapping	Each captured drop shows the boundary to snap to and the gap: "1:16 (−2.2s)", or "on the 8" in green when it lands square (drops are held in whole seconds, so anything inside that half-second of rounding counts as square). A drop sitting well off the grid is telling you that moment is not phrase-aligned — choose deliberately rather than assume.
First downbeat at	The grid assumes bar 1 starts at 0:00, which is wrong for tracks with an intro. Set this field (per song) and the whole grid shifts. It travels in the drops library and the transfer codes, so it is set once per song, not once per week.
Song-relative	Drops are stored as positions inside the song — durable, reusable, unaffected by playlist position or crossfade.

On a short screen REV 20	The sheet never grows taller than the screen; if its content does not fit it scrolls inside itself. The title row and the × stay pinned at the top while you scroll, and a full-width **Done** button sits at the foot. This matters in split screen — at half height the sheet used to overflow upward and take the × off the screen with it.

Colour-coding the drops REV 19

Tap a drop chip and it steps to the next colour in the key. Seven taps takes it all the way round and back to plain, so a mis-tap is undone by carrying on tapping. The chip shows the time as before, tinted — no words are added, and nothing about the colours reaches the Build Sheet or the Cue Notes.

The key	Six colours sit above the chips with their names, shown once a song has drops. Defaults: Intro · Build · Drop · Chorus · Hook · Breakdown.
Edit key	Rename any entry and recolour it from the phone's colour picker. **Reset** returns all six names and colours to the originals.
Restyling repaints	A drop stores a slot number, not a colour. Recolour slot 1 and every drop coded slot 1 changes with it, across every song.
Very dark colours	A near-black choice would vanish against the dark screen, so the editor names any entry at risk. It does not stop you.
Where it travels	Backups, transfer codes and the drops library. The Build Sheet and the Drops Library Editor carry the coding through untouched without displaying it.

Adding your own keys REV 35

Add key	In **Edit key**, beside Reset. It appends a slot with a fresh colour and puts the cursor in its name, ready to type. Up to fourteen in total; the app ships with six.
Removing one	A key you added carries a ×; the six the app ships with do not. Removing one renumbers the keys above it — in the playlist *and* in the stored drops library — so no drop is left pointing at a key that moved. Drops coded with the key you removed become plain again, and the prompt counts them first.
Reset	Puts the original six names and colours back, exactly as shipped. Because that shortens the key, any drop coded with a key you added becomes plain; the prompt says how many before you commit.
Tapping through	Tapping a chip cycles it through every key you have, not just the first six, then back to plain.
Between devices	The key lives on the device; the drops carry slot numbers. A drop coded with a key a second device has not got shows plain there rather than half-painted — set the same key up on both, or Reset both, and they agree again.

Bands, defaults, transfer and backup

Bands & RPM settings	Every block's BPM range and ride mode, plus the universal RPM cap. Save as my defaults sets your baseline; Reset to my defaults returns to it; Factory reset restores the originals.

Send to sheet / Import code	A transfer code carrying playlist, BPMs, blocks, per-song modes, drops, colour coding, song lengths, band settings and saved defaults. Paste a Build Sheet code in by mistake and it tells you which button to use instead.
Backup / Restore	Downloads or reloads a dated .json of everything. Works after the fact — the old copy of the tool need not still exist.
Where a saved file goes REV 22	There is no separate Save to… button any more — every Backup, Export file, Save file and Export CSV button asks where the file should go. On a phone that is the share sheet, so it can go to Drive, Files or mail; on a computer it is the ordinary save dialog, so it lands in a folder you pick. A browser that can do neither still downloads as before. See Section 7.

3 · THE BUILD SHEET

Structure

Factory layout: Warm-Up · Segment 1 Rolling Opener (endurance) · Recovery Bridge · Segment 2 The Grinder (climb) · Recovery Bridge · Segment 3 The Challenge (sprint) · Segment 4 Finisher · Cool-Down. Segment numbers renumber themselves to match ride order. **Add section** creates a named custom section; each header's × removes one; **Reset to defaults** always restores the factory eight, and the library and drills survive it.

Renaming a section REV 26	The title in a coloured header is a field. Click it and type; the songs in that section do not move. The working note underneath is editable the same way. Both print as plain text — no boxes, no outlines.
Switching two sections REV 26	The ⇄ menu in a header trades that section's whole identity — title, note and colour — with another section's. Every song stays exactly where it sits. This is the opposite of ▲▼ and the position dropdown, which move a section *with* its songs; use ⇄ when the music is right and the label is wrong.
What stays put REV 26	Clock windows (“10:00 – ~24:00”) describe the slot in the ride, not the block, so they never move with a title. Neither do the per-row role tags that **Import ← Selector** matches on — a renamed section still takes the same kind of song, because the roles belong to the rows.
Getting the stock title back REV 26	Once you rename or switch a section, this file stops refreshing its title, note and colour from the template — your wording is yours. The ⇄ menu's ↺ **Back to the stock title** hands it back and restores the refresh. Clearing a title and clicking away does the same. A section you never touched keeps picking up template improvements as before.

FTP and Max HR REV 24

Where	The header line under the title: **Rider: FTP [158] W · Max HR [164] bpm**. Both are boxes you can type in. They print as plain text, so a printed sheet still reads as a sentence.

What they drive	The whole zone table underneath. Power columns are shares of FTP — Z1 under 55% · Z2 56–75 · Z3 76–90 · Sweet Spot 88–94 · Z4 95–105 · Z5 106 and up — and heart-rate columns are shares of maximum: Z1 under 60% · Z2 60–70 · Z3 70–80 · Sweet Spot 76–84 · Z4 80–90 · Z5 90–100. The table redraws as you type.
Where they live	On the device, like the drift setting, and they travel inside sheet backups and Send → Build Sheet codes. **Reset to defaults** rebuilds the template but leaves your numbers alone — they are yours, not part of the ride.
Bad input	Clearing a box or typing nonsense puts the shipped default back when you click away. There is no way to leave the table broken.
Retesting REV 33	An FTP test changes one number and the zone table follows at once. The per-song Power and HR cells are free text — your written cues — so nothing rewrites them behind your back; **Refresh targets** in the toolbar rescales them when you are ready.
Refresh targets REV 33	Rescales every Power and HR cell that still holds the wording the template shipped with, which was written for FTP 158 W and Max HR 164. “Z4 · 150–166 W” at FTP 200 becomes “Z4 · 190–210 W”. Zone labels keep their own digits — Z1→Z2 and Z5+ are never mistaken for numbers — and a target that is words rather than watts, like “max safe”, is left alone. It scales from the original shipped figures every time, so pressing it at one FTP and then another does not compound rounding: set 158 again and you get the original wording back exactly.
What it will not touch REV 33	Any cell you have typed yourself. Once a cell is yours it stays yours, through any number of refreshes. The prompt counts those rows before you commit, shows an example of the change, and Cancel does nothing. The rescaled rows are tinted for a few seconds so you can see what moved.

Per-song rows

#	Continuous song count across the sheet — also the Spotify playlist position.
Song / Intervals / Drops / Power / HR	Wrapping fields; long text grows the row rather than truncating.
Len	Song length (m:ss); auto-fills on import from a CSV, and from the Selector when the song carries one. Feeds all clock math. A row that has a song but no length is tinted amber until you fill it — on screen only, never on the printout. REV 19
BPM → Ride → RPM	Type the BPM, pick 1:1 or ½; RPM computes under the 110 cap. The beat-lock cannot be broken here.
Pos	Seat / Stand / S-St.
Σ ride REV 43	Where that song's block *ends* on MOWL's timeline — which is where the next song starts, crossfade included. Read it against the Session Designer's song strip, row for row. The last song's figure is MOWL's total for the class; its audio keeps playing for another 10 s underneath whatever follows. Until s60-v49 this column reported where each song's audio ended instead, so every row read 10 s per transition adrift of the timeline.

◉ ride / ◉ 8-bar	Under the drops: each drop as an absolute class minute (drift-corrected), then the 8-bar phrase boundary it snaps to and the gap — "1:16 −2.2".

Drill presets and the drill catalog

Each Intervals cell has a ▾ drills dropdown. Picking a drill fills the cell in one tap, and the text stays fully editable afterward — a preset is a starting point, not a lock.

The list itself	Four groups — Steady/Tempo, Climb, Sprint, Recovery — plus *My drills* once you add your own.
— clear this cell —	Empties the Intervals cell. The way back to blank.
▾ Save this row's text as a drill	Shelves whatever you have typed into the My drills group, so a wording you invented becomes reusable. It will not save a duplicate.
✎ Manage drills…	Opens the catalog panel (below).
What a drill is	Always a resistance, position or effort pattern keyed to song structure — a chorus, a drop, a verse. Never a cadence change, because cadence is set by the beat-lock.

The Manage drills panel

Opens from the ▾ drills dropdown. The whole catalog — built-in drills and your own — is editable here, in two sections: the **groups**, then the **drills** inside them.

Groups

Rename	Type over a group's name and every drill in it follows. The starting names — Steady/Tempo, Climb, Sprint, Recovery, My drills — are only a suggestion.
Reorder	The ▲ on each row moves a group up, which changes the order the groups appear in the dropdown.
Delete	The × removes a group. Its drills move to another group rather than disappearing, and the confirm tells you how many will move.
＋ Add group	Names a new one. New drills you save land in the LAST group, so keep your catch-all at the bottom.
Drill count	Each row shows how many drills it holds, so an empty group is obvious.

The drill catalogue in Excel REV 31

Export to Excel	In **Manage drills**. Writes `Saturday60-Drills-YYYY-MM-DD.xlsx` — a real workbook, not a CSV, that opens without a warning. No internet needed: the app builds the file itself, so it works with no signal like everything else here.
The layout	Two sheets mirroring the panel. **Groups** — Order, Group — one row per heading in the order the app lists them. **Drills** — Group, Drill — one row per drill, which is exactly the pair of controls on each row of the panel. Both have a frozen header, a filter and sized columns. A third sheet, **Read me**, states the rules; the importer ignores it.

The Group dropdown REV 32	Click any cell in the Drills sheet's Group column and Excel offers a dropdown listing the groups from the Groups sheet — nothing to type, and no drill stranded under a misspelled heading. Excel refuses anything not on the list and says why. Add a group on the Groups sheet and it joins the dropdown at once; there is room for twenty more than you currently have, so the last few entries in the list may be blank until you use them.
Import from Excel	Reads a .xlsx back. It replaces **every** group and drill on that device, so export first if you want a copy of what is there. The prompt counts the drills and groups, names any group the Drills sheet invented, and says how many empty rows it skipped. Cancel changes nothing. Drill text already typed into the Build Sheet is never touched — this is the catalogue, not the ride.
Renaming a group	Change it on **both** sheets. Matching is by the name spelled in the Drills column, so renaming only on the Groups sheet gives you an empty new heading and leaves the drills under the old one. Find and Replace down the Drills column is the quick way. The Order column is what sets the order — re-sort in Excel and the app follows.
What it will not read	.xls, .csv, or a Numbers file. It says so rather than failing quietly. The sheets must still be called Groups and Drills.

Drills

Reword	Type straight into a drill's text box. Changes save as you type and appear in every Intervals dropdown.
Regroup	Each row has a group selector, listing whatever groups you currently have.
Delete	The × removes a drill, built-in ones included.
+ Add drill	Creates a blank entry to type into directly, without needing a row on the sheet to save from.
Restore built-ins	Puts back any original drills you have deleted — and any default groups you have removed — while keeping everything you have added. Never creates duplicates.
Done	Closes the panel and refreshes every dropdown on the sheet.

Groups are your own filing system: they do NOT map to anything in the workbook. Excel's Segment column comes from the Build Sheet's section, role and position; a drill's TEXT goes to Cue Notes and its group never leaves the dropdown. Rename them to suit how you think.

Both the groups and the drills are stored on the device and shared by the two Build Sheet variants. They ride inside sheet transfer codes and sheet backups, and MERGE rather than replacing on import — new items are added, nothing you have is wiped, and no drill is ever left in a group that does not exist. A drill also travels with its song in the drops library, so a song's wording comes back with its drop times.

Reordering

Songs: ▲▼ move within a section and across boundaries; the per-row section dropdown teleports a song. **Sections:** a position dropdown (1–8) plus ▲▼. Clock windows stay attached to positions, so the timeline stays ascending no matter what moves.

Totals and buttons

Header tallies	"3 songs · Σ 13:36 · thru 24:07" — count, section length, cumulative ride time; +n? flags missing lengths.

Grand bar	The whole playlist against 60:00, crossfades included, green under / red over.
Print / Save as PDF	Clean printout; edit chrome hidden, colours and wrapped text kept, amber length tints suppressed.
Import Spotify export	A Chosic/Exportify CSV or pasted text (Section 4).
Import ← Selector	The Selector's playlist by feed or code; fills matching slots by block role. Lengths sent from the Selector are written in; where a slot changes to a different song and no length was sent, the old runtime is cleared rather than left attached to the wrong track, and the box is flagged amber. REV 19
Import drops only	Merges ONLY the Drop times, matching songs by title. Everything else untouched; it reports any titles it could not match.
Send → Selector	Packages the sheet's songs, BPMs, ride modes and drops as a Selector code.
Send / Import ← Build Sheet	Whole-sheet clone, sheet to sheet only. Buttons name their destination so the two code types cannot be confused.
Drops library	The shared song → drops (and drill) file. **Fill from library** inside it fills this sheet from what the device already knows; Add this sheet's drops banks the week. See Section 4. REV 21
Phrase map	Every song's 8/16/32-bar boundaries converted to the ride clock, so an interval in Session Designer can start and end on the music. Needs only BPM and Len — no drops required. A **match MOWL / real audio** switch decides whether drift is applied; a per-song offset field handles tracks with an intro. Copy as text puts the whole map on the clipboard.
Export for Excel	Copies the sheet as tab-separated rows for the workbook (Section 5).
Backup / Restore	A dated .json of the whole sheet, Section Library, drill catalog and drift.
Save (header) + Section Library	Shelve a section as a named block; the bottom panel loads blocks into any section by Replace or Append, or deletes them.
Drift __ s/song	Fractional-seconds correction applied to ◉ ride times only (Section 6).
Reset / Clear	Reset to defaults restores the factory 8-section sheet; Clear songs & drops keeps your targets.

Hiding the targets REV 34

The switch	**Hide targets** in the toolbar drops the Power target and HR guide columns; it then reads **Show targets** to put them back. The choice is remembered on that device, on screen and on the printout.
Nothing is lost	The wording stays on every row, hidden rather than deleted, and comes back exactly as you left it. Imports, backups and transfer codes carry it either way, so a sheet sent with the targets hidden arrives complete.
What the width buys	Song goes from 138 to 212 pixels and Intervals from 122 to 182, so long titles and long drill wording stop wrapping. That is the whole point of the layout: on the bike you are reading the music and the drill, not the watts.

History	This replaces the separate *Build Sheet · no targets* file that predates the app. That file never received any of the app's work — no drops colour coding, no song length, no Fill from library — and it kept its own storage, so the two sheets drifted apart. A switch cannot.

4 · HOW EVERYTHING CONNECTS

Data flow: Spotify → Exportify or Chosic CSV → Build Sheet → (printed PDF · Excel workbook · MOWL Session Designer), with the phone Selector feeding drops, BPMs and song lengths in both directions.

Which export REV 43	**Exportify is the one to use when the clock matters.** Its Duration (ms) column carries the exact length in milliseconds, which is what lets the sheet round the way MOWL does. Chosic's Duration is mm:ss with the fraction already thrown away, so the sheet has to add a second back blindly — right for almost every track, and one second over for a track whose true length is an exact whole second. Chosic still wins on Camelot and Explicit, so exporting both and using Chosic for the audit is reasonable.
Chosic CSV carries	Song, Artist, BPM, Duration (mm:ss) — plus Camelot (key matching) and Explicit (clean audit).
Exportify CSV carries REV 43	Track Name, Artist Name(s), Tempo (BPM) and Duration (ms) — exact lengths, no Camelot or Explicit.
Spotify-export import	Fills top-to-bottom in ride order, bridges included: titles, artists, BPMs, lengths. Over 110 BPM becomes ½ automatically. Playlist order = sheet order.
Lengths round UP REV 43	MOWL counts each track rounded up to the next whole second, so the sheet stores it that way: an Exportify duration of 189.363 s becomes **3:10**, and a Chosic **3:09** has a second added. That means the Len column reads one second longer than the same song in MOWL's own song list — that is not a mismatch, it is what makes the timeline agree. Verified against a 23-song class: rounding up and subtracting the crossfade gives 1:01:43, exactly MOWL's total. Rounding down gives 1:01:31 and rounding to nearest 1:01:39; neither fits.
Selector ↔ Sheet	Selector code → sheet fills by block role. Sheet's Send → Selector maps sections back to blocks (Segment 4 → Finisher; bridges and cool-down → Cool-Down) and carries drops and ride modes both ways. Song lengths travel Selector → Sheet.
Sheet ↔ Sheet	Sheet codes clone everything. Sheet content REPLACES on import; the Section Library and drill catalog MERGE.
Two import routes	CSV route = fast skeleton with BPMs/lengths. Selector route = complete data including drops, colour coding and any lengths you set. Pick ONE per week.
Seeing that it landed REV 29	When an import runs, the rows it wrote are tinted blue for a few seconds and the first of them is scrolled into view, so the result is visible wherever you were on the sheet. The toolbar line reports the counts for eight seconds. If an import changes nothing at all — the same file imported twice — it says so outright rather than looking like a dead button.

An import replaces the playlist REV 30	Importing a Spotify or Chosic export fills the sheet in ride order and then puts the sheet back to its normal shape: every slot past the last imported song is emptied, and any **spare rows** beyond a section's factory count are removed outright — so a short playlist gives a short sheet, not a tail of blank rows. A section never drops below its factory row count (one row for a section you added yourself), so the ride structure survives. The prompt says how many slots will be cleared and how many rows removed before you commit, and Cancel leaves the sheet untouched. More songs than slots still extends the cool-down, as it always has — and the next smaller import shrinks it back.
Sheet → MOWL REV 43	Σ gives each song's start on the Session Designer's song strip, and the last song's Σ is MOWL's total — checked song by song against a real session. From an Exportify CSV it lands on the second; from Chosic expect the odd song a second out, for the rounding reason above. Set interval counts from the Intervals column, nudge boundaries to Drop times, set targets from Power (Z2 ≈ 56–75% FTP · Z3 ≈76–90% · SS ≈88–94% · Z4 ≈95–105% · Z5 ≈106%+).

Getting drops into the sheet — which route

Three routes exist and they overlap. The choice depends on where the ride was built:

Sheet built from a CSV	**Import drops only.** In the Selector tap Send to sheet, move the code across, then paste it into the Import drops box. It fills the Drop times column and nothing else — BPM, ride mode, position, intervals and targets are left exactly as they are.
Ride built on the phone	**Import ← Selector.** Same code, different button: it fills songs, BPMs, ride modes, drops and lengths into matching slots by block role.
Different devices, or for reuse	**Drops library.** Export the .json from one tool and import it in the other. Slower, but it accumulates week to week and also carries the drill, the downbeat offset and the colour coding.
Both tools on one device	Skip the transfer — the clipboard code goes straight from one to the other, and both tools now read the same storage.
Do not do both	Import drops only and Import ← Selector read the SAME code. Run one or the other, not both.

The drops library — moving it between the tools

Both tools read and write the same file, **Saturday60-Drops-YYYY-MM-DD.json**. Since the tools moved to one web address they also share one stored library on a given device, so the file is only needed to move between devices or to keep an archive. An entry holds the song, its drop times, its colour coding and its drill.

Fill from library REV 21	In **both** tools, inside the Drops library panel. It looks up every song — the Selector's playlist, or the Build Sheet's rows — and fills whatever that song has not got yet, then reports the split: "Filled 4 songs from the library. 2 songs already had their drops. 1 song is not in the library yet." The Selector fills drops, colour coding and the downbeat offset; the Build Sheet fills drop times, the drill and the downbeat. Nothing already filled is touched. Adding a song does not fill it automatically — this button is the trigger.

Phone → desktop	In the Selector: Drops library → Add this playlist → Export file. Move the file across. In the Build Sheet: Drops library → Import file. Every matching song gets its drops (and drill) filled in.
Desktop → phone	The same in reverse: Add this sheet's drops → Export file → Import file in the Selector.
The one rule	**Load before you Save.** Import the file into a tool before exporting from it. Each export writes that tool's own accumulated library, so exporting from a copy that was never loaded would produce a thinner file than the one you already have.
What it will not overwrite	Only songs with NO drops yet are filled; anything already marked is left alone. The report gives the split.
Matching	By title, with the artist suffix, capitalisation and punctuation ignored — "Titanium" and "Titanium — David Guetta, Sia" are the same song. A different edit in the title — "(Radio Edit)" — is a different song to the matcher.
Editing outside the tools	The Drops Library Editor opens the file directly: times shown in m:ss, drills alongside, add or delete songs, merge two libraries (richer entry wins), then save back out. Renaming a title rebuilds its matching key automatically. Colour coding is preserved through the round trip without being shown. **Export CSV** writes Saturday60-Drops.csv — Song, Artist, BPM, Drops, Drill — as a copy to read or print; the tools still read only the .json, so keep using Save file for anything that has to travel. REV 21

5 · THE SPIN CLASS PRO WORKBOOK

Purpose: **Charts** gives a visual read of the ride's shape, and **Playlist Print** is the paper fold-up backup for when MOWL malfunctions or isn't available.

Getting the ride in

Build Sheet → **Export for Excel**, then click **B2 on My Playlist** and paste. The export carries no quote characters — Excel's clipboard parser treats quotes as text qualifiers and would mangle anything containing them, so even the Elapsed formula is written quote-free.

A · #	Not pasted — numbers itself, blank on empty rows.
B · Song Name	Title only; the artist is split off at the " — " separator.
C · Artist	The other half of that split.
D · Runtime	Zero-padded mm:ss, right-aligned.
E · Elapsed Time REV 43	A live formula, pasted with the block; empty rows show blank. It subtracts the crossfade once per song, so it reads the same figure as the Build Sheet's Σ — where the next song starts on MOWL's timeline. The crossfade it uses is **N2** on My Playlist (default 10), labelled in N1; that is the only place to change it.
F · RPM	The beat-locked cadence.
G · Gear (M3) / H · Gear (M3i)	Blanked on paste — re-pick from the 1–24 and 1–72 dropdowns.

I · Segment	Warm-up · Fast-Flat · Seated/Standing-Climb (by position) · Sprint/Standing-Attack (by position) · Recovery (bridges) · Cool-down.
J · Cue Notes	Intervals text, then " | drops 1:05, 2:47" when the row has drop times. Colour coding is never written here.
K · Pointers	NEVER pasted over — yours for coaching phrases.
L · Avg RPM	NEVER pasted over — weighted average from the Intervals sheet; falls back to column F.
M · _TimeValue	Hidden helper formula; not pasted.

The paste block is B:J. If a column ever moves, the export's Elapsed formula must be re-checked against the sheet — that is the one place the two files have to agree exactly.

The Intervals sheet

Eight slots per song, each a **seconds + RPM** pair, plus Total sec, **Avg RPM** (time-weighted), Song sec and **Covered**. Leave a song blank and it uses its headline RPM.

One slot per cadence	Order and repetition don't matter — only total seconds at each RPM. A Tabata of 8 × (20s on / 10s off) is two slots: 160s at the work cadence, 80s at the recovery cadence.
What it drives	Avg RPM feeds the M3 / M3i Load Curves, and each interval's seconds land in its own cadence band.
Keyed to row position	Intervals belong to a row, not a title. New week, new songs — clear them (ClearPlaylist does).

Cadence bands — not power zones

Time in Cadence Bands (Low under 70 · Mid 70–89 · High 90+) measures *legs turning, not watts*, and is named that way to avoid confusion with the true power zones on the Build Sheet. A light-geared song can sit in the High band while barely taxing you.

A watts-per-interval version with real Z1–Z5 and a planned TSS is possible but not built: it roughly doubles data entry, and normalized power computed from interval blocks overstates short efforts because it skips the 30-second smoothing real NP uses. Any such figure would be planned load, not what the bike measures.

Playlist Print — two sides

Page 1	Ride detail, three lines per song: title (with Pointer), a stats strip — RPM, both gears, runtime, elapsed, segment — then the cue.
Page 2	Song and artist in larger type. Print double-sided, flip on the LONG edge, and the same songs land on the same half front and back.
Geometry	0.25in margin | 3.66in panel | 0.60in gutter | 3.66in panel | 0.25in margin. The 4.25in fold falls inside the empty gutter. Top/bottom 0.75in, header/footer 0.3in. Content sits just under the page so fit-to-page never shrinks it — a shrink would slide the fold off position.
Empty slots	Dropped entirely. 24 songs prints at about 13pt titles, 19 at about 16pt, 12 at the 26pt ceiling.

How many fit REV 43	Up to **32**, matching the rows ClearPlaylist and RepairFormulas work on. The older module counted only rows 2–25 and capped at 24, so a longer class lost its tail off the printout without saying so. Type shrinks as the count rises, down to a 10pt floor.

The three macros

Install (once)	Save as .xlsm, enable the Developer tab, Visual Basic → remove any older SaturdaySixty module → File › Import File… → the .bas. Close and save.
Weekly rhythm	**ClearPlaylist** → paste at B2 → pick the gears → **FormatPlaylistPrint** → print.
1 · ClearPlaylist	Wipes only last week: paste block (B2:J33), Pointers (K2:K33), intervals (Intervals C2:R33) — all keyed to row position. Formulas, charts, drills and the drops library untouched. Runs RepairFormulas silently.
2 · FormatPlaylistPrint	Counts songs, splits them evenly, drops empty slots, scales rows and fonts, builds both pages, resets page breaks, sets margins and print area.
3 · RepairFormulas	Rebuilds A, E, L, M on My Playlist and the Intervals formulas, and seeds the crossfade setting in N1/N2 if it is missing (never overwriting a figure you have set). Columns A, L and M sit OUTSIDE the paste block, so if a row loses them they never return on their own — while E is inside and gets rewritten by every paste. The result is an Elapsed formula with no input: no error, just a blank cell on that row.
Why the .bas is ASCII	The VBA editor imports as Windows-1252; a UTF-8 dash arrives as strange characters, so special characters are built with ChrW().
Elapsed and the crossfade REV 43	Elapsed now subtracts the crossfade, so the workbook, the Build Sheet and MOWL all give the same figure. It did not before: its total ran 10 s per transition long — about 3:40 on a 23-song class. Both sides had to change, because the Build Sheet's **Export for Excel** writes column E and would otherwise put the old formula straight back over a repaired sheet. Update the .bas *and* the Build Sheet together, or the bug returns on the next paste. Charts and Intervals were never affected — they read Avg RPM, gears and per-song runtime, not Elapsed.

6 · THE TIMING AND PHRASE MATH

Crossfade

With a 10 s crossfade, real clock time = sum of lengths −10 s per transition. A 16-song hour plays ~2:30 faster than its raw sum. All Σ figures use this, and MOWL was verified to use the same convention.

The lengths it works from are **rounded up** to the whole second, because that is how MOWL counts (Section 4). The two together are what make the sheet and the Session Designer agree: a 23-song class summing to 1:01:43 in MOWL comes out at 1:01:43 on the sheet. Get either half wrong and the error compounds a second or so per song, which looks like a small discrepancy on song 2 and a minute by the finisher.

Song-relative vs ride-clock drops

Stored drop times are positions *inside the song*. The ◉ ride line converts them to class minutes: song start = Σ −length, drop = start + offset.

Fractional drift

Tracks carry fractional seconds that whole-second interval maps truncate, so music falls ~0.4–0.5 s further behind the intervals each transition — measured at 6.5 s over 15 transitions. The **Drift s/song** field corrects ◉ times only; Σ stays matched to MOWL. It **starts at 0.43**, the measured figure, so a sheet is correct from the first song you type and nothing has to be ridden first; set it to 0 to show MOWL's own clock. Recalibrate: gap at the last song ÷ transitions. In MOWL, stretching every other interval by +1 s holds the hour within ~half a second.

Setting it REV 43	**Set…** beside the Drift box opens a panel that leads with the typical figure and a **Use typical 0.43 s/song** button, together with what that adds up to across the songs on the sheet. That is the whole job while you are building: measuring from a ride is offered underneath as an optional refinement, not a prerequisite. The button was labelled **Measure…** before s60-v49 and the panel opened on step 1 of a measurement, which read as an instruction to ride the class before the sheet would give you numbers.
Refining it from a ride REV 40	The Build Sheet does the arithmetic. The panel asks two things: how many seconds behind the music was when the last song started, and how many songs were in that ride — which it fills in from the sheet. Drift is the gap divided by the transitions before it, so a 6.5-second gap over a 16-song ride is 6.5 ÷ 15 = **0.43**. **Use this number** writes it into the Drift box.
Reading the gap	Once you do want to refine it: ride or watch the class with the sheet in front of you and look once, at the moment the last song starts: where the music actually is against where the sheet says it should be. One reading at the end is more accurate than several along the way, because the error accumulates and a bigger number is easier to read.
Sanity checks	The panel refuses a negative gap — music running ahead is not drift — and warns when the answer exceeds 3 s per song, which usually means a song was skipped rather than that the clock slipped. It also shows the total the figure implies at the end and at halfway, so you can tell at a glance whether it matches what you saw.
How often	Once, then again if you change platform, bike or crossfade setting. It is a property of how the music is cut and played, not of a particular week's playlist.

Bars and phrase boundaries

Assuming 4/4: one beat = 60 ÷ BPM, one bar = 240 ÷ BPM. At 126 BPM a bar is 1.90 s, 8 bars 15.2 s, 32 bars 61.0 s. The tools work in **boundary times** rather than bar numbers, because what you need is where a drill can start, not where you happen to be.

8 is the working grid	Dance and pop is built in 8-bar sections, with 16 and 32 as the bigger groupings. Most drops land on an 8; 32-bar alignment is much rarer, since a 32-bar phrase runs about a minute and a 3½-minute song holds only three or four.
When a drop misses the 8	Every drop off by the same amount → the downbeat offset is wrong, not the music (by far the most common cause). One drop off by a beat → you marked the impact, not the downbeat. Off by 2 or 4 bars from a point onward → the song has an odd-length section, common in remixes and sped-up edits. Scattered fractions → stopwatch noise; a waveform editor removes it.

What actually matters	It matters less that the drop sits on a boundary than that your effort STARTS on one. Begin a sprint at the 8-bar mark before the drop and the drop lands inside the effort — riders get a bar or two to load up rather than reacting cold.

The phrase map — boundaries on the ride clock

The Build Sheet's **Phrase map** converts every song's boundaries into class time using the same crossfade and drift maths as the ◉ ride column, so a boundary can be typed straight into Session Designer. It needs only BPM and Len, so a ride is programmable from the CSV alone — drops become refinement rather than a prerequisite.

match MOWL (default)	Ignores drift. These times line up with Session Designer.
real audio	Applies your drift setting. These match what the room hears, and run later than MOWL's clock — at 1.0 s/song the third song is already 2 s adrift, which is how a mismatch usually announces itself.
Per-song offset	A small field on each row of the map, for tracks that open before bar 1.

This is the arithmetic half of what a DJ phrase counter does. The other half — finding the true first downbeat and confirming the tempo holds — needs the audio file, and is worth doing in a waveform tool for the songs where phrasing really matters; read the downbeat there and type it into the offset field.

7 · DATA, DEVICES AND BACKUP

Where data lives REV 19	In the browser's storage on the device where you typed it, keyed to the web address. All three tools now share one address, so they share one storage area on that device — the Selector's drops library is the Build Sheet's drops library. Nothing is uploaded; there is no account and no server.
Per device, still	Storage does not follow you between phone and laptop. Backups and transfer codes are how work moves.
Data from the old loose files REV 19	A browser kept separate storage for every file path, so nothing saved in the standalone HTML files carries over. Open the old file, press **Backup**, then press **Restore** in the app. Do it for the Selector and the Build Sheet once.
Backup files (best)	Each tool's Backup button writes a dated .json; Restore reads it back on any device at any time, and unlike clipboard codes it works AFTER the fact. The drops library exports the same way.
Saving to the cloud, and .txt REV 22	On a phone the save buttons hand the file to the share sheet, because that is the only route to Drive, Files or mail — the phone's own file picker sees local storage only, and Chrome on Android offers no download-folder choice on a handset with no SD card. Phone share sheets refuse .json on security grounds, so the shared copy is named .txt — identical contents, and Restore and Import accept either. On a computer the same buttons open a real save dialog instead.
Transfer codes (quick)	Selector ↔ Selector, Sheet ↔ Sheet, and Sheet → Selector. Snapshot, not live sync — the device you Send from wins. The paste box accepts codes of any size and cleans line breaks from email wrapping.

What a full backup is	A Selector backup + a sheet backup + the drops library: playlist, drops, colour coding, song lengths, bands, defaults, ride modes, the whole sheet, the Section Library, the drill catalog and the drift calibration.
When to back up	Monthly, and always before uninstalling the app or clearing browsing data. Removing an installed app can clear its storage on some phones.

8 · TROUBLESHOOTING

Import looks like it did nothing REV 41	First check the toolbar line at the top of the sheet and look for the blue-tinted rows; from s60-v33 the sheet scrolls to the first imported row, so this usually answers itself. If the prompt appears, you press OK and the sheet genuinely does not change, that is a fault in the app rather than in your file. It happened up to s60-v46 when an import carried **more songs than the sheet had slots** and the sheet had no cool-down section — deleted, renamed away, or replaced by sections of your own. The extras had nowhere to go, the import stopped mid-way without a word, and the panel stayed open. Fixed: extras now land in the last section, whatever it is called, and any other failure says so instead of stopping silently.
Closing a panel shows another REV 42	Each panel — Drops library, Spotify import, Phrase map, the transfer code, Manage drills, the drift calculator — used to open on top of whatever was already open, so closing the top one uncovered the one still sitting underneath. From s60-v48 opening any panel closes the rest, and **Esc** closes whatever is showing.
Cancel where Import should be REV 41	On a phone the buttons at the foot of an import panel used to wrap so the last one sat on top of the one before it, and a tap aimed at **Import** could land on **Cancel**. Fixed in s60-v47; the buttons now share the width evenly on a narrow screen.
Import left last week's songs behind REV 28	Fixed in s60-v32. Before that, an import only overwrote as many slots as it had songs and left the rest alone, so a short playlist landed on top of a full sheet and the two mixed. Update, then import again.
Import says "No songs found"	The CSV header wasn't recognised. The importer knows Chosic (Song/Artist/BPM/Duration) and Exportify (Track Name/Artist Name(s)/Duration (ms)).
"That code could not be read"	Usually the tail was cut in transit, or it's the wrong code type. Re-copy with the whole block selected; compare the last 10 characters on both devices.
An Elapsed Time is blank	Run RepairFormulas. See Section 5.
Elapsed runs ahead of MOWL REV 43	The old formula is back, which means a paste from an out-of-date Build Sheet put it there. Update both the Build Sheet and the .bas module, then run RepairFormulas or re-export and paste again. Check E2: the current formula contains SUMPRODUCT and reads N2.
Σ is a second or two out per song REV 43	The lengths came in from Chosic, whose mm:ss has already lost the fraction. Re-import from Exportify, whose Duration (ms) is exact. See Section 4.
An upload doesn't show up REV 43	Bump VERSION in sw.js, then close every tab and the installed app and reopen. From s60-v50 the worker calls skipWaiting and fetches pages network-first, so a single reload is enough; before that a new version sat waiting behind any open tab. Incognito is the quick way to see what the server is really holding.

Three print pages instead of two	Stale page breaks from an earlier run. Fixed — FormatPlaylistPrint now resets them every time.
A BPM looks doubled or halved	Spotify-derived tempo data does this. Tap-test and edit the BPM.
Times don't match a database	Databases (and AI memory) get durations wrong; the CSV is the referee — Exportify for the exact figure, Chosic for a whole-second one.
Key/Camelot codes	Key detection runs well under 50% accurate on some services. Match within segments only; bridges are free reset points.
Explicit versions	Spotify search and Spotlistr default to the ▪ version. Audit the Chosic Explicit column.
Smart Reorder	Premium, mobile-only. Useful within a block; it will sand off the deliberate tempo cliffs between blocks.
Intervals drift from the music	Section 6. Growing drift = fractional seconds (set Drift); constant steps = crossfade mismatch; irregular = Automix snuck on.
Boundaries look wrong	The track opens with an intro — set the downbeat offset for that song.
Phrase map disagrees with MOWL	The map is on the real-audio clock. Switch it to match MOWL, and check the drift value is your measured one (0.43), not a guess.
Only nine phrase boundaries listed	Working as designed: the grid runs eight phrases past the downbeat and grows as you mark drops. Switch to 32 bars to see further ahead.
Lengths went blank after an import REV 19	A slot that changed to a different song loses the old song's runtime, because it would otherwise corrupt the ride clock. Amber boxes mark what to fill. Set lengths in the Selector and they arrive with the songs instead.
Cannot see the × on the drops screen REV 20	Fixed in s60-v13. Before that the sheet had no height limit, so on a half-height window (split screen) its top scrolled off and the close button went with it. The × is now pinned and there is a Done button at the bottom. If you are still stuck on an old version, the back gesture or turning the phone landscape gets you out.
Drops did not fill in REV 19	Adding a song does not consult the library. Open Drops library → **Fill from library**. If a song is still skipped, its title differs from the stored one by more than the artist suffix.
Excel: run-time error 429	VBA cannot create ThisWorkbook — a fault in the Office install, not the workbook. Use the primary macro module, which avoids ThisWorkbook entirely; keep only one copy of the workbook open when running macros. Quick Repair then Online Repair (Settings › Apps › Microsoft 365 › Modify) fixes it properly; a later 365 update may too.
Sections look scrambled	Reset to defaults is always one button home.

9 · THE APP ONLINE — INSTALLING, UPDATING, ROLLING BACK

The three HTML tools are published as one installable web app at **`https://indoor-cycling.github.io/saturday60/`**. Opening that address gives a hub screen with the three tools on it. The whole app

is stored on the device the first time you visit, so it runs with no signal at all — a basement studio or a phone in aeroplane mode makes no difference.

The hub REV 38 lists the three tools as cards, then the three documents — Weekly Workflow, System Manual, Getting Started — as one short **Documents** group, each with a PDF link beside it and **Copy the link to share** underneath. They were three full cards until s60-v42, which pushed the install panel most of a screen further down on a phone.

Installing · iPhone / iPad	Open the address **in Safari** (Chrome on iOS cannot do this). Share button → Add to Home Screen → Add. A "Sat 60" icon appears with your other apps.
Installing · Android / desktop	The hub's **Install on this device** button appears when the browser is ready. Otherwise: Chrome ⋮ → Add to Home screen, or the install icon at the end of a desktop address bar.
Do you have to install it?	No. It works the same in an ordinary tab, offline included. Installing gives it its own window and icon, and browsers protect an installed app's storage more carefully. Both read the same data on that device.
Which version am I on?	The hub shows it — "Version s60-v12" — with a **Check for updates** button beside it. Or open `…/saturday60/sw.js` and read the VERSION line near the top.

The files in the repository are the only source of truth REV 21

A copy in a downloads folder, attached to an old message, or pasted into a chat weeks ago is not. Before asking anyone — or any AI assistant — to revise a tool, fetch the current file from the repository and hand over *that*. Every tool carries shared machinery an older copy will silently lack: the app shell (`app.js` , the manifest and icons, which provide the back-to-hub bar, the save-destination prompts and the offline install), the drops colour coding, and the song length that travels into the Build Sheet. A revision built from an earlier copy removes those without saying so, and the loss shows up later as missing colours or a page that no longer works offline. The repository's `README.md` says this, and every file opens with a comment saying the same, so the warning travels with the file even when nobody thinks to mention it.

Publishing a change

The app is hosted from a GitHub repository. Editing it is uploading files.

1 · Replace the files	In the repository: **Add file** → **Upload files**, drag in what changed, **Commit changes**. A matching filename overwrites; nothing duplicates. When in doubt, upload the whole folder — it costs one extra drag and removes any doubt about which version is live.
2 · Bump the version	Open `sw.js` and raise the number on the `VERSION` line near the top: `var VERSION = 's60-v13';` . That one word is what tells installed copies there is something new. Forget it and the update still lands, just a launch or two later.
3 · Take it on the phone	Wait about a minute for GitHub to republish, open the app, and tap **Update now** on the bar that slides up. Your songs, drops and sheets live on the phone and are never touched by an update.
If no bar appears	Check the repository finished publishing (green tick), then open …/sw.js and confirm the VERSION line near the top shows the new number, then tap Check for updates, then close the app fully and reopen it.

Rolling back

A version number only ever goes forward. To undo a change you upload the older files under a *higher* version number — re-uploading an older number would leave every installed copy unaware anything had changed.

How	Upload the previous files with `sw.js` stamped higher than the current one, for example `'s60-v13-rollback'`. Commit, wait, tap Update now.
Your data	Untouched. A feature that stored something extra leaves that data dormant rather than deleting it, and rolling forward again brings it back.
Keep the zips	The simplest safety net is keeping the last known-good folder somewhere. There is no version history to fall back on beyond what GitHub itself records.

The rollback bundle REV 45

There is **one**, and it holds the version you are running now. After every successful update, `Current_Revisions_Cloud/saturday60-rollback-current/` is refreshed with a complete copy of the files just published, and its `sw.js` is stamped `s60-vNN-restore`. Whatever is live has been used and is working, so a copy of it is kept ready; if the next update misbehaves you upload that folder and you are back on solid ground. It is not in the repository — a bundle stored there would be overwritten by the very upload it exists to undo.

Complete, not a patch	Every file, including `app.js`, the manifest and the icons, not only the ones that happened to change. That is what makes a half state impossible: anything the broken update added that is not in the folder is replaced.
One step back	It goes back one version, not many. For anything older the dated folders beside it are archives — `2026-10-05_rollback-to-v48` returns the app to its state before the October work — and GitHub holds the full history of every file besides.
Refresh it after each update	Replace the contents with the new live files and move the stamp to match. A bundle three versions stale undoes three things you wanted.
The Excel module rides along	The Elapsed formula is written from both sides — the Build Sheet's Export for Excel and the module's RepairFormulas — so `SaturdaySixty.bas` sits in the bundle too. It is imported in Excel rather than uploaded to GitHub, and the note says so.
Earlier practice	Bundles used to be made per change, each holding only the files that change touched. It worked, but every bundle needed a version stamp clearing the next release, and the stamps collided as soon as two bundles existed. One complete bundle removes the arithmetic.

Why the version line only has to differ REV 45

The browser compares the **bytes** of `sw.js` against the copy it already holds. It never reads the number. Tested against a miniature of this worker: a *lower* number, a same-number `-rollback` suffix and a higher number all installed correctly, and the old cache was purged every time. The one thing that does nothing is re-uploading a byte-for-byte identical `sw.js` — which is exactly what happens if you re-upload an old version folder verbatim, and is the real meaning of "a version number only goes forward". The `-restore` suffix exists so that cannot happen.

Privacy

The repository is public, which is what free hosting requires — so the tools' code is readable by anyone with the address. Your content is not: songs, BPMs, drop times, colour coding and build sheets are written to storage on the

device you typed them on and never leave it. The app has no server and asks for no account.

Quick values: crossfade 10 s · drift 0.43 s/song · RPM cap 110 · FTP 158 W and Max HR 164 (both editable) · TSS ~65–70 · bar = 240 ÷ BPM · current app version s60-v60.

The app: `https://indoor-cycling.github.io/saturday60/` — index.html · selector.html · buildsheet.html · drops.html · Saturday60_WeeklyWorkflow.html and .pdf · FTP_5min_Test.html · RPE_Watts_Card_Editable.html · RPE_Watts_Card_YMCA_Editable.html · app.js · sw.js · manifest.webmanifest · icons/ · README.md. Alongside it: Spin_Class_ProV4 workbook · SaturdaySixty_PlaylistPrint.bas · Saturday-60_Weekly-Workflow.pdf. When any tool changes, ask Claude for a manual revision to match.

10 · POWER & TESTING

These three tools make the ride power-based. Both the Keiser M3 (the YMCA bikes) and the M3i Studio Plus at home read Watts; once a rider has an **FTP** — the power they can hold for about an hour — every effort can be named in Watts, not only by feel. The **5-Minute FTP Test** finds that number; the two **RPE + Watts** cards turn it into zones and gear/cadence starting points. All three are their own pages in the app, opened from the hub under **Power & testing**, and work offline.

The 5-Minute FTP Test

A guided version of Keiser's own five-minute Personal Power Test, 30 minutes end to end. The page runs a timer that names the phase, counts it down and flips the surge/ease cues for you, beeps at each change and at the start of the main effort, and keeps the screen awake. A mute toggle silences the cues.

Warm-Up 1 · 5 min	80–90 RPM · 70–120 W · HR 60–70% of max. Easy spin to open the legs.
Warm-Up 2 · 10 min	Add 1–2 gears, then keep that gear. Alternate 1 min at 100 RPM and 1 min at 80–90 RPM, five times. HR 75–80%.
Main Test · 5 min	The highest power you can hold for the full five minutes — even pace, strong form. 80–90 RPM, HR > 85%. Record the average Watts at the end.
Cool-Down · 10 min	Ease the gears down, power under 100 W, 70–80 RPM. HR 60–65%.

The result goes through one sum: `FTP = 5-min average − 15% = average × 0.85`. Keiser's worked example of a 250 W average gives an FTP of 212 W. The 15% is the correction for a five-minute test; a shorter effort needs a bigger one and a longer effort a smaller one — **3-min −20% · 5-min −15% · 10-min −10% · 20-min −5%**.

Enter the five-minute average in the calculator on the page and it does the sum; two buttons, **Open my M3i card** and **Open YMCA card**, then carry that FTP straight into a card (as `?ftp=` on the link), in the same window.

On an M3, reset the display before the test and stop pedalling for three seconds at the end to read the flashing averages. On an M3i Studio Plus, start with the time counter at 00:00.

RPE + Watts — the two cards

Each card takes an FTP (and a personal-best Watts) and lays the seven power zones out as a percentage of that FTP, next to the RPE 1–10 feel scale and a **gear × cadence** cheat — the same Watts reached either by a heavier gear turned slower or a lighter gear spun faster. Type a new FTP and every number re-computes; print at 100%.

M3i card (home)	Gears out of 72, for the Studio Plus. Two panels — the RPE + Watts scale and the gear cheat.
YMCA card (class)	Gears out of 24, for the M3. Three panels — a **Find Your Number** benchmark, the scale, and the gear cheat. An optional rider name prints on each panel, so riders print their own.

The zone edges are fixed shares of FTP — Z1 under 55% up to Z7 above 150%, with the top of Z7 set by the personal best. If a raised FTP pushes 150% past the stored best, Z7 shows open-ended (for example `319W+`) as a nudge to retest the best with a short all-out effort. Re-test FTP every four to six weeks and reprint.

The M3's Watts are displayed but not lab-calibrated the way the M3i's are, so treat a rider's Watts as a measure against themselves over time, not an exact number to compare bike to bike.

How the three connect

Find the number in the FTP Test, tap through to a card and it opens with the FTP already filled in; the two cards also link to each other. Nothing is uploaded — the FTP you type lives only for that visit, and the pages run offline once the app is installed.

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